

Autofocus-Based Optimization of Trajectory Filtering for Close-Range ISAR Imaging

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Abstract—Radar-based imaging of the plantar foot surface enables contactless monitoring of deformation and swelling during gait. This work presents an inverse synthetic aperture radar (ISAR) imaging approach that integrates trajectory estimation from an Azure Kinect depth camera. As the raw depth estimates contain noise and jitter, a filtering and optimization framework is introduced to enhance reconstruction quality. A discrete Kalman filter with constant-acceleration dynamics is optimized using autofocus metrics that quantify image homogeneity and edge sharpness. The feedback loop automatically tunes filter parameters based on the reconstructed image quality rather than trajectory error. Experimental results with a moving 3D-printed foot model demonstrate that the proposed method significantly improves focus and surface consistency compared to unfiltered or root-mean-square-error (RMSE)-based filtering. These findings establish a foundation for contactless, radar-based medical imaging of the plantar foot in dynamic conditions.

Index Terms—ISAR, tracking, autofocus, close-range, medical imaging.

I. INTRODUCTION

Conventional imaging modalities such as ultrasound and magnetic resonance imaging (MRI) can visualize internal plantar structures, including fascia and soft-tissue edema [1]. However, ultrasound requires direct skin contact and MRI is restricted to static, high-cost examinations, making both unsuitable for dynamic or contactless surface imaging. Surface-based optical measurements, on the other hand, have already been shown to capture clinically relevant swelling information [2], motivating radar-based methods for contactless plantar surface imaging.

Recent advances in close-range MIMO radar imaging have shown that high-resolution tracking of human extremities is feasible in biomedical contexts [3]. Complementary work by Brinkmann et al. [4] demonstrated similar close-range imaging capabilities for large technical objects, confirming the general applicability and scalability of MIMO-ISAR techniques. Together, these studies underscore the potential of close-range radar methods for precise surface reconstruction across different domains.

Building on these concepts, this work investigates ISAR-based imaging of the plantar foot surface using trajectory estimates obtained from an Azure Kinect depth camera. As illustrated in Fig. 1, a moving foot above a uniform linear array is tracked by the depth sensor to synthesize an aperture

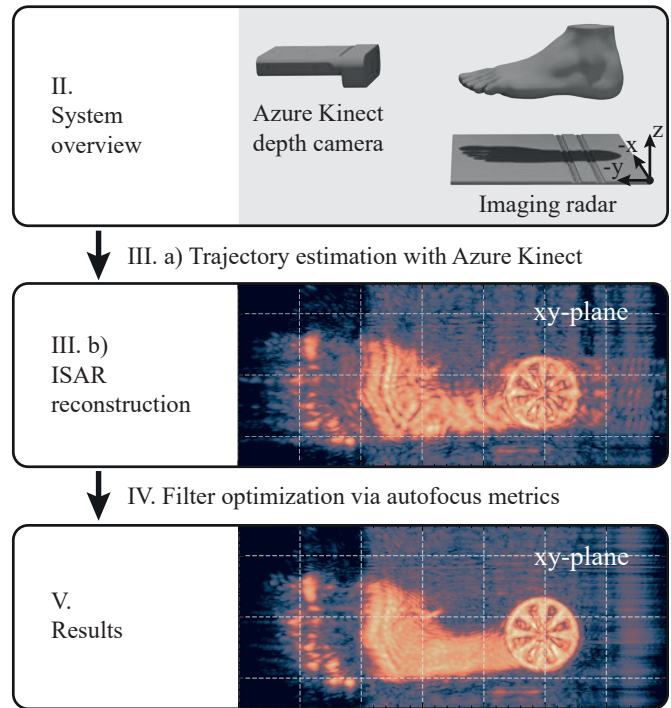


Fig. 1: A moving foot above a uniform linear array is recorded using an Azure Kinect depth camera to estimate its trajectory. These estimates are used to synthesize an aperture for ISAR reconstruction and plantar foot surface imaging. A tracking filter is optimized via autofocus metrics to enhance the reconstruction quality.

for ISAR reconstruction. To enhance image focus and surface fidelity, tracking filters are optimized via autofocus metrics, resulting in improved reconstruction quality even under non-linear motion conditions.

The remainder of this paper is organized as follows. Section II introduces the experimental setup and gives a system overview. Section III describes the trajectory estimation with the Azure Kinect and the ISAR reconstruction procedure. Section IV details the proposed optimization framework and the autofocus metrics used for the feedback loop. Section V presents the reconstruction results and Section VI concludes the paper with a summary and outlook on future work.

II. SYSTEM OVERVIEW

The experimental setup used in this work is illustrated in Fig. 2. A 3D-printed foot model is mounted on a linear stage

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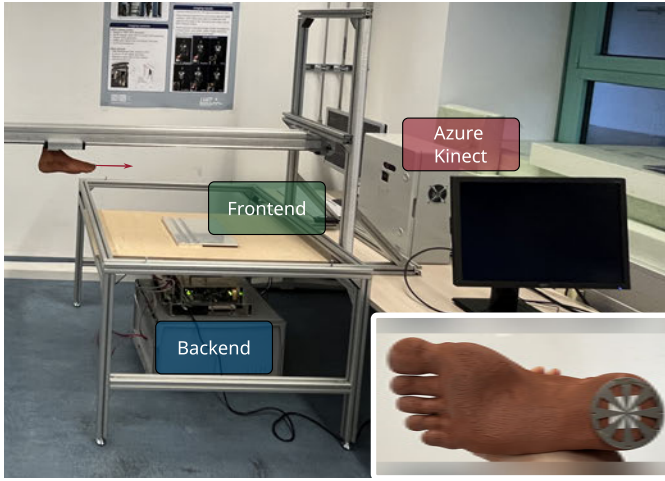


Fig. 2: Measurement setup. A 3D-printed foot model is mounted on a linear stage, facing an Azure Kinect, and moving above the radar. The position estimates from the depth sensor are used for ISAR reconstruction of the foot. A close-up view of the plantar side of the foot is included for reference.

that moves above the imaging radar, thereby synthesizing an inverse aperture for ISAR reconstruction. An Azure Kinect depth camera is positioned in front of the model to continuously estimate the foot's 3D position relative to the radar.

The radar module is embedded in a table that can be equipped with a transparent plexiglass top to enable real-world scenarios with walking humans. A close-up image of the bottom side of the foot model is shown in the lower right corner of Fig. 2 for visual reference. A Siemens star is added to the heel, for better visualization of the performance enhancement and measure of reconstruction quality. To emulate the specular reflection characteristics of human skin in the millimeter-wave regime [5], the surface of the model was coated with copper spray. The Azure Kinect sensor is connected to the radar backend and synchronized via Precision Time Protocol (PTP). The linear stage itself, however, operates independently and is triggered manually.

The antenna layout of the radar is depicted in Fig. 3. It consists of a linear array of transmit (Tx) antennas (red dots) and a parallel line of receive (Rx) antennas (blue dots) spaced by approximately 41 mm. Each line forms a uniform linear array with an element spacing of 3 mm along the x -axis. Since

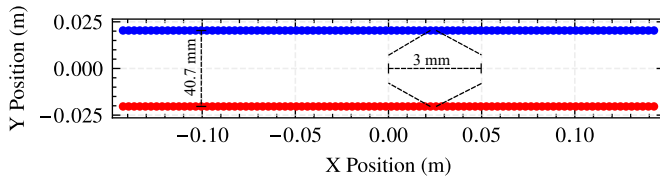


Fig. 3: The antenna layout of the imaging radar: the blue dots represent the receiving antennas, and the red dots represent the transmitting antennas.

the antennas are implemented as open waveguides, the system provides no inherent focusing in the direction of motion; the

Tab. I: Radar Parameters

Parameter	Value
Maximum frequency	80 GHz
Bandwidth	10 GHz
Number of frequency steps	64
Frame duration	6.5 ms
Range resolution	1.5 cm
Maximum unambiguous range	0.94 m
Num Tx antennas	94
Num Rx antennas	94
Frames for reconstruction	380
Cross-range resolution δ_x	1.8 mm

synthetic aperture is therefore generated solely through the object's translation. The Cartesian cross-range resolution δ_x at the center of the array in the distance L can be calculated as

$$\delta_x = \frac{c_0}{4f_{\max}} \sqrt{4\left(\frac{L}{D}\right)^2 + 1}, \quad (1)$$

where D denotes the array width, $f_{\max}$ the maximum radar frequency and c_0 the speed of light in vacuum [5]. The same formula holds also for the Cartesian cross range resolution in y -direction δ_y , where the array width D is now the distance of the sampled aperture that the object spans over the radar.

The radar operates as a stepped-frequency continuous-wave (SFCW) system with the parameters listed in Tab. I. A total of 64 frequency steps are employed to synthesize a bandwidth of 10 GHz, thus creating a range resolution of about 1.5 cm. With $D \approx 0.30$ m and $L = 0.25$ m, the resulting cross-range resolution in the foot plane is approximately 1.8 mm. Due to a frame rate limit of 154 Hz, the maximum translation velocity of the foot during measurements has to be constrained accordingly to avoid aliasing artifacts.

III. TRAJECTORY ESTIMATION AND ISAR RECONSTRUCTION

A. Trajectory Estimation with the Azure Kinect

Two motion profiles were considered to evaluate the system performance. First, a constant-velocity trajectory (CV) was employed; subsequently, a constant-acceleration motion (CA) was designed to more closely resemble the swing phase of a human walk. In both cases, the 3D-printed foot model was initially positioned 1.4 m away from the Azure Kinect. It then moved along a trajectory that ended at 1.0 m, resulting in a total travel distance of 0.4 m. Due to hardware constraints, only 380 radar frames could be captured, resulting in a total recording length of approximately 2.5 s and an effective aperture length of about 25 cm used for ISAR imaging. For the constant-velocity case, the foot moved at $v = 100$ mm/s. In the constant-acceleration scenario, the model accelerated for 1.5 s until reaching the same velocity, maintained that speed for 2.5 s, and then decelerated back to rest over another 1.5 s.

The 3D trajectory was estimated using the Azure Kinect depth camera. All pixels associated with the foot were extracted using the DBSCAN clustering algorithm, and the

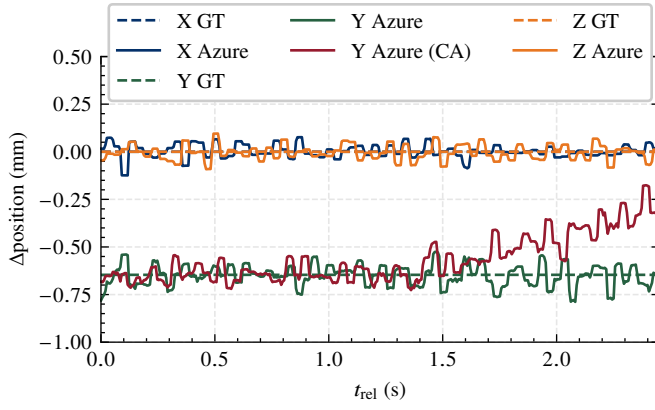


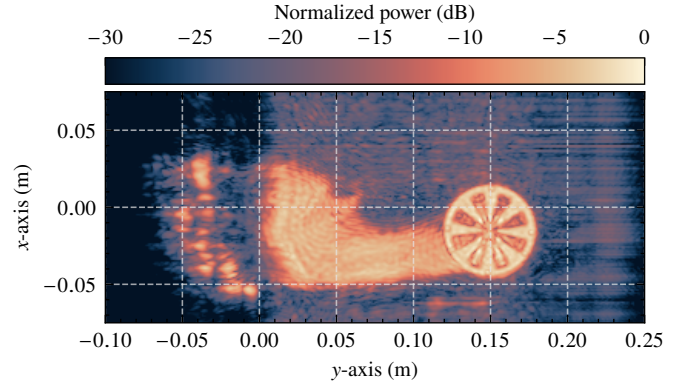
Fig. 4: Frame-to-frame 3D displacement derived from Azure Kinect estimates, temporally interpolated to match the radar frame rate. The ground-truth translation (dashed lines) and the Azure Kinect estimates (solid lines) are shown for the x, y, and z directions in the constant-velocity measurement case. For the constant-acceleration (CA) case, only the Azure Kinect trajectory estimate in the moving direction is displayed.

centroid of the resulting cluster provided the instantaneous position estimate. The estimated positions were transformed into the radar coordinate system, interpolated to match the radar's sampling rate and used for ISAR reconstruction. Fig. 4 illustrates the relative translation of the foot over the radar frames, comparing the ground truth (dashed) with the Kinect estimates (solid). Because the linear stage was triggered manually, ground-truth data were unavailable for the acceleration case; however, the estimated trend remains visible in the red curve of Fig. 4.

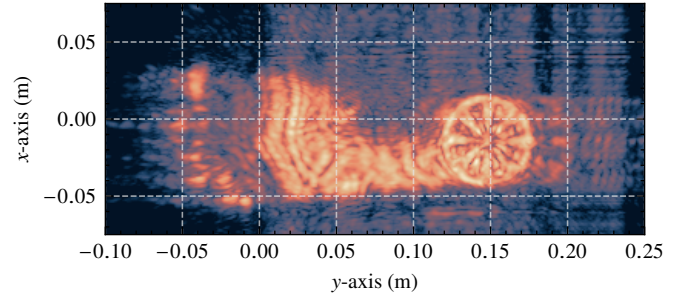
B. ISAR Reconstruction

For image reconstruction, the backprojection algorithm from [6] was applied due to its robustness against nonlinear motion trajectories and optimal reconstruction quality within close-range conditions. Fig. 5 shows the resulting maximum intensity projections (MIPs) of the reconstructed 3D radar volumes: Fig. 5a presents the reconstruction based on the ground-truth motion from the linear stage, while Figs. 5b and 5c show the reconstructions using unfiltered Azure Kinect position data for the constant-velocity and constant-acceleration cases, respectively.

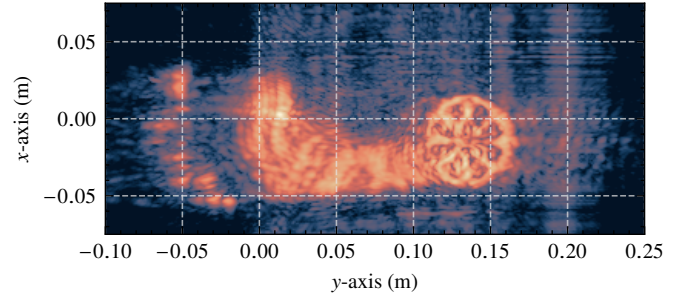
Compared to the ground-truth reconstruction, the images obtained from the Kinect-based motion estimates show noticeable degradation. The previously homogeneous surface of the foot is distorted, now showing local maxima and minima. Especially in the Siemens-star region, these distortions result in defocusing and deformation of the object, particularly for the accelerated motion case. The observed artifacts are primarily caused by noise in the estimated trajectory, underlining the need for proper trajectory filtering to enhance ISAR reconstruction quality.



(a) Ground-truth ISAR reconstruction for the constant-velocity trajectory.



(b) ISAR Reconstruction using unfiltered Azure Kinect depth estimates for the constant-velocity case.



(c) ISAR using unfiltered Azure Kinect depth estimates for the constant-acceleration case.

Fig. 5: ISAR reconstructions of the foot for the constant-velocity (b) and constant-acceleration (c) motion using unfiltered Azure Kinect depth estimates. Subfig. (a) shows the reference ground-truth reconstruction for the constant velocity case.

IV. FILTER OPTIMIZATION VIA AUTOFOCUS METRICS

Selecting an appropriate trajectory filter for ISAR reconstruction is not straightforward, as the optimal configuration depends on several unknown factors, such as the noise characteristics of the position estimates and the motion dynamics of the target. For example, a trajectory with constant velocity motion can be smoothed more aggressively compared to an accelerated motion. Additionally, the measurement noise may affect the ISAR reconstruction differently in each spatial direction. It is therefore difficult to predict how trajectory errors will propagate through the backprojection algorithm and

degrade the reconstruction quality.

To address this challenge, we propose an optimization loop that employs autofocus metrics as feedback criteria, as illustrated in Fig. 6. The 3D position estimates from the depth sensor are filtered using candidate parameters provided by the Optuna optimization framework [7]. For trajectory filtering, a Kalman filter based on a constant-acceleration motion model was employed [8]. The state vector comprises position, velocity, and acceleration in three dimensions, while assuming additive Gaussian process and measurement noise.

Process noise is introduced at the acceleration level to capture deviations from the constant-acceleration assumption. It is modeled as zero-mean Gaussian with variance σ_a^2 , identical across all spatial axes. The resulting discrete-time process covariance matrix is constructed as

$$\mathbf{Q} = \mathbf{G} \sigma_a^2 \mathbf{I}_3 \mathbf{G}^\top, \quad (2)$$

where $\mathbf{G}$ contains the Δt -dependent integration terms of the constant-acceleration model. This formulation ensures physically consistent propagation of uncertainty from acceleration to velocity and position. Measurement noise is modeled as zero-mean Gaussian with covariance

$$\mathbf{R} = \sigma_z^2 \mathbf{I}_3, \quad (3)$$

where σ_z^2 denotes the variance of the position measurements provided by the depth sensor. Similar to the process noise, the measurement noise is assumed identically across all spatial axes to limit the dimensionality of the optimization problem. These parameters control the trade-off between trajectory smoothness and responsiveness to measurement updates.

For the constant velocity trajectory, a Rauch-Tung-Striebel (RTS) smoother [9] was additionally applied to refine the filtered trajectory. For the constant-acceleration case, the RTS smoother was omitted, as the fixed-interval smoothing attenuated valid acceleration dynamics, leading to motion-induced reconstruction artifacts despite a reduction in speckle-like fluctuations. Instead, a standalone Kalman filter was used. For each candidate, the filtered trajectory is interpolated to match the radar's frame rate, and ISAR reconstruction is performed. The resulting image quality is then quantified using autofocus metrics. The optimization process is terminated using an early-stopping criterion based on the relative improvement of the objective metric. Let J^* denote the best metric value observed so far. A new trial is considered an improvement only if it exceeds the current best value by at least a relative factor ϵ , i.e.,

$$J_{\text{new}} > J^*(1 + \epsilon). \quad (4)$$

If no such improvement is observed for N consecutive trials, the optimization is terminated. In this work we set the $\epsilon = 10^{-3}$ and $N = 50$. Since the focus of this work is to showcase the potential of autofocus-driven optimization rather than to minimize computational cost, the stopping parameters were chosen to allow sufficient convergence of the objective metric. The computational bottleneck is the backprojection reconstruction, while the Kalman filtering and

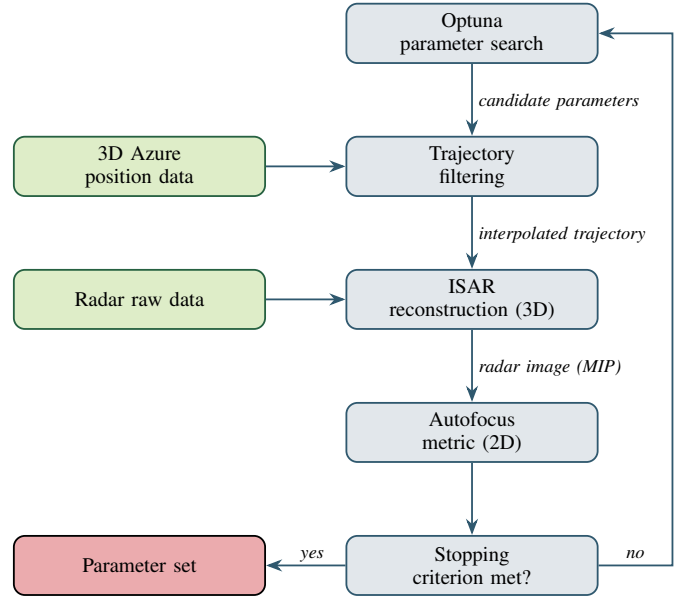


Fig. 6: Feedback loop for optimizing the filtered trajectory of the foot via autofocus metrics. The 3D position estimates from the Azure Kinect are filtered with candidate parameters provided by Optuna's optimization framework. An ISAR reconstruction is then performed and the resulting image is evaluated with the proposed autofocus metrics. The loop is continued as long as a significant improvement within the metrics can be observed.

metric evaluation contribute only marginal overhead. During the optimization runs, the objective typically exhibited an initial increase followed by a saturation period.

The choice of the autofocus metric depends on the imaging target and the desired image characteristics. In this work, we evaluate two complementary metrics that emphasize different aspects of the reconstructed image: (1) patch-wise weighted homogeneity, which measures smoothness of the foot's plantar surface, and (2) weighted edge density, which highlights structural sharpness around regions like the Siemens star. For the patch-wise weighted homogeneity metric, the 2D maximum-intensity-projection (MIP) image is divided into non-overlapping patches of $m \times n$ pixels (e.g., 10×10). For each patch P_k , the gray-level co-occurrence matrix (GLCM) homogeneity $H(P_k)$ from [10], is computed and weighted by the mean intensity $I(P_k)$ to emphasize regions belonging to the imaged subject. The overall metric H_{pw} is then defined as

$$H_{\text{pw}} = \frac{1}{M \cdot N} \sum_{P_k} (m \cdot n \cdot H(P_k) \cdot I(P_k)), \quad (5)$$

where M and N denote the image dimensions. An example of the spatial distribution of the homogeneity metric H_{pw} is shown in Fig. 7, overlaid on the reconstructed MIP of the foot. As the Siemens star would interfere with the evaluation of this metric for the later application in a real-world scenario, only the cut out part of the foot that is displayed in Fig. 7 was used for optimization with this metric. As can be seen, it strongly emphasizes regions with both high reflection strength and homogeneous amplitude distribution.

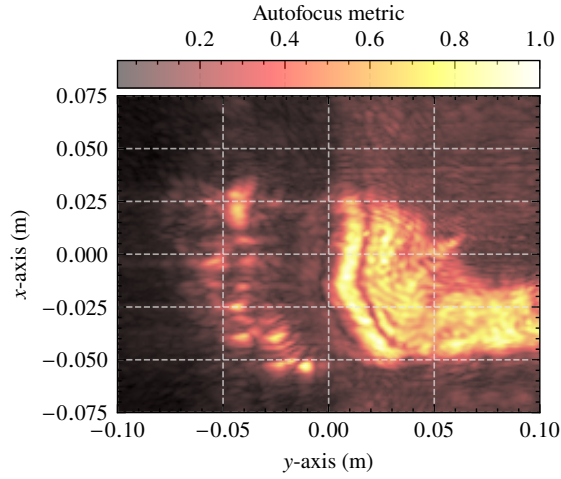


Fig. 7: Exemplary MIP of the foot's surface overlaid with the spatial distribution of the homogeneity-based autofocus metric. The Siemens star region was excluded in the optimization process, as it will not be present in real-world measurements.

The second metric, the weighted edge density D_w , is based on the Sobel gradient magnitude [11]. It quantifies the proportion of strong edges weighted by pixel brightness and is defined as

$$D_w = \frac{\sum_{i,j} |S(i,j)| I(i,j)}{\sum_{i,j} I(i,j)}, \quad (6)$$

where $S(i,j)$ denotes the normalized Sobel response and τ is a predefined threshold (e.g., 0.2). The normalization by the total image intensity ensures that edges occurring in brighter regions contribute more strongly to the overall metric. This metric is particularly sensitive to structural sharpness and is therefore strongly influenced by high-contrast features such as the Siemens star. In contrast to the homogeneity-based metric H_{pw} , which evaluates surface consistency, D_w emphasizes edge definition and provides a complementary autofocus criterion. For brevity, the detailed visualization of this autofocus metric is not included here.

V. RESULTS

To evaluate the performance of the proposed optimization framework, several optimizer runs were conducted and compared against the ground-truth reconstructions. Additionally, the results were verified against a reference filter tuned on a noisy, randomly generated trajectory by the means of minimizing the root-mean-square error (RMSE) of the position estimates. A summary of all reconstruction results is presented in Fig. 8.

The ground-truth reconstruction, representing the ideal image quality, is shown in Fig. 8a. The upper row (Figs. 8b and 8c) depicts the results for the constant-velocity (CV) trajectory, while the lower row (Figs. 8d to 8f) illustrates the reconstructions for the constant-acceleration (CA) case. To enable quantitative comparison, the three-dimensional volume of the Siemens star region from the ground-truth image

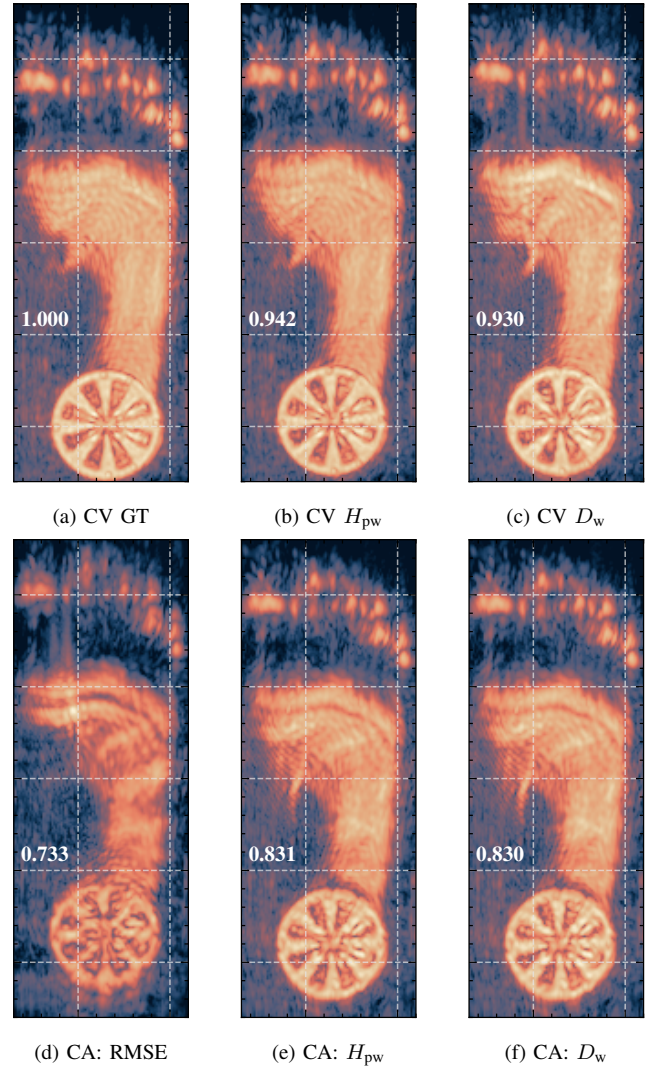


Fig. 8: Results of the optimization using different autofocus metrics. The upper row shows the constant-velocity (CV) case: (a) ground-truth reconstruction, (b) reconstruction optimized using the weighted patch-wise homogeneity metric H_{pw} , and (c) reconstruction optimized using the weighted edge density metric D_w . The bottom row shows the constant-acceleration (CA) case: (d) reconstruction using a reference filter optimized on trajectory minimizing RMSE, (e) optimization using weighted patch-wise homogeneity H_{pw} , and (f) optimization using edge density D_w .

was extracted and complex-correlated with the corresponding regions in the reconstructed images. The resulting correlation coefficient is displayed in each subfigure. Since there is no ground truth reconstruction for the CA case, the Siemens star from the CV case is also exploited here for the correlation calculation. Thus a perfect correlation value of 1 cannot be achieved and the correlation values are expected to be lower than in the constant-velocity case. Additionally, since a constant deceleration was performed during the later part of the trajectory, the overall aperture length—and thus the achievable resolution—differs from the constant-velocity case. However, as the physical object remains identical, the

relative trend of the correlation values between the different metrics remains meaningful. Two main observations can be made from these results. First, both proposed autofocus-based optimization metrics—the weighted patch-wise homogeneity H_{pw} and the weighted edge density D_w —produce reconstructions with strong spatial coherence and high structural detail, particularly within the Siemens star region. For the constant-velocity case, the homogeneity metric achieves a correlation value of 0.942, whereas the weighted edge-density yields 0.930. In comparison, the unfiltered CV reconstruction reaches only a correlation value 0.822, demonstrating the quantitative gain achieved through autofocus-based optimization. The main difference between both autofocus metrics can be observed in the structure of the right side of the Siemens star, where the homogeneity-based optimization more closely resembles the ground-truth Siemens star. Second, the nearly identical correlation values obtained with both metrics in the CA case suggest that the achieved reconstruction quality represents a practical upper limit for this filter type in combination with Azure Kinect-based trajectory estimation. Additionally, to put the optimization with the proposed autofocus metrics into perspective, Fig. 8d is included, where the optimization objective was to minimize the RMSE. As can be seen, both the image quality and the correlation coefficient of the Siemens star are significantly degraded.

Even though the homogeneity metric H_{pw} was applied only to the image region excluding the Siemens star, it achieves better performance compared to a metric specifically designed to target the known structure of the Siemens star. This showcases the potential of this metric for real-world use cases, where no such calibration pattern is present.

VI. CONCLUSION

In this work, we laid the foundation for radar-based medical imaging of the plantar side of the foot using an Azure Kinect depth camera for ISAR reconstruction. Because noisy position estimates led to defocused images, we proposed an optimization loop based on autofocus metrics to enhance reconstruction quality. The results demonstrate improved image focus and surface homogeneity, even for trajectories with constant acceleration, indicating the potential of the method for contactless medical foot imaging. Further work will focus on the real trajectory of a human foot, incorporating a dynamic movement with rotation in all three spatial dimensions. Since this work solely relies on the estimates of the Azure Kinect depth camera, further investigations will also implement tracking information of the radar itself, as first investigations in [12] have shown to look promising.

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